Sobha Realty Delayed Transaction & Revenue Exposure Analysis

Final Capstone Report

Group Number: 08
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1. Executive Summary

This project looks at delayed property transactions at Sobha Realty using SQL. About 1 in 4 transactions in the 2018-2023 data are delayed, and the delay rate nearly tripled over that period, from 11% to 33%. Sobha Creek Vistas carries the biggest risk, with 1,809 delayed transactions and close to AED 1.59 billion tied up. Sobha Hartland - The Crest is the opposite case, the highest-selling project with zero delays. The report ends with five recommendations for where management should look first.

2. Problem Statement

Sobha Realty has a large portfolio of property transactions across multiple projects and property types. A significant number of transactions in the available 2018-2023 dataset are classified as delayed. This project looks at where these delayed transactions are concentrated, works out the revenue tied up in them, and compares delay rates across projects to find areas that may need more management attention. The analysis only covers patterns in the transaction data and does not try to explain why a delay happened.

Stakeholder

Sobha Realty's management and sales operations team. They would use this to decide which projects, property types, and buyer segments need closer monitoring.

3. Dataset and Data Quality

The dataset came from an actual client business (kept confidential here). It covers property transactions from 2018 to 2023, originally 105,283 rows across 116 projects. Since this is private client data and not publicly licensed, it is used here only for this academic analysis and the raw file is not published.

Cleaning steps

- Removed duplicate rows
- Removed rows with missing or invalid project names
- Removed transactions with a price of zero or blank
- Removed personal buyer/seller details (name, mobile number, ID numbers) for privacy
- Kept only Buyer-side rows, so each transaction is not counted twice

After cleaning, 46,763 transactions were used in the analysis.

Data Dictionary

Column	Meaning	Type	Example	Notes
Regis	Date the transaction was registered	Date, mixed formats	28/12/2023	Two formats found in the file, handled with TRY_STRPTIME
Project	Name of the project	Text	Sobha Hartland Waves	Some rows had '0' as a placeholder, those rows were removed
PropertyTypeEn	Type of property	Text	Flat	A few rows are blank
ProcedureValue	Transaction value in AED	Stored as text	1700000	Converted with TRY_CAST, rows with 0 or blank removed

Column	Meaning	Type	Example	Notes
ProcedureNameEn	Transaction status	Text	Delayed Sell	Used to identify delayed transactions
ProcedurePartyTypeNameEn	Buyer or Seller side of the row	Text	Buyer	Filtered to Buyer only, to avoid counting each transaction twice
Size	Property size in sqft	Numeric	1250	Used in Milestone 2 to group into size categories
CountryNameEn	Buyer/seller nationality	Text	India	Used in Milestone 2 nationality analysis

4. Main and Supporting Questions

Main question

Where are Sobha Realty's transaction delays concentrated, and which projects, property types, and buyer groups carry the highest risk?

Supporting questions

1. What is the total transaction volume?

2. What percentage of transactions are delayed?

3. How have delays changed year by year?

4. Which projects have the most delays?

5. Which property types are affected the most?

6. Which projects have the highest revenue exposure?

7. Which project and property type combinations are the riskiest?

8. Which project has high volume and a low delay rate?

9. Does property size affect delay rate?

10. Which buyer nationalities are most affected?

11. Which projects have repeated delays year after year?

12. Which property type is growing the fastest?

13. What are the highest-value delayed transactions?

5. SQL Methodology

All queries were run in DataLab, using DuckDB. Techniques used: SELECT, WHERE, GROUP BY, ORDER BY, COUNT, SUM, AVG, CASE WHEN (both to flag delayed transactions and to build size categories), TRY_CAST and TRY_STRPTIME (to clean price and date fields that weren't stored as proper numbers or dates), CTEs for the multi-step year-by-year check, and HAVING to drop low-volume groups before ranking. A JOIN wasn't possible since the data sits in a single table (the second sheet in the original file was empty), so a CTE was used instead where multi-step logic was needed.

6. Results Summary

Metric	**Value**
Total buyer transactions	46,763
Delayed transactions	12,683 (27.12%)
Delay rate in 2023	33%
Most delayed project	Sobha Creek Vistas, 1,809 delayed
Highest delay rate by property type	General Use, 99.6%
Average delayed transaction value	AED 2,011,760
Sobha Creek Vistas revenue exposure	AED 1.59 Billion
Best performing project	Sobha Hartland - The Crest, 3,472 transactions, 0% delay
Highest delay rate by size	Medium, 800-1500 sqft, 53.34%
China buyer delay rate	54.35%, on 2,769 transactions
Highest single delayed transaction	AED 95.6 Million

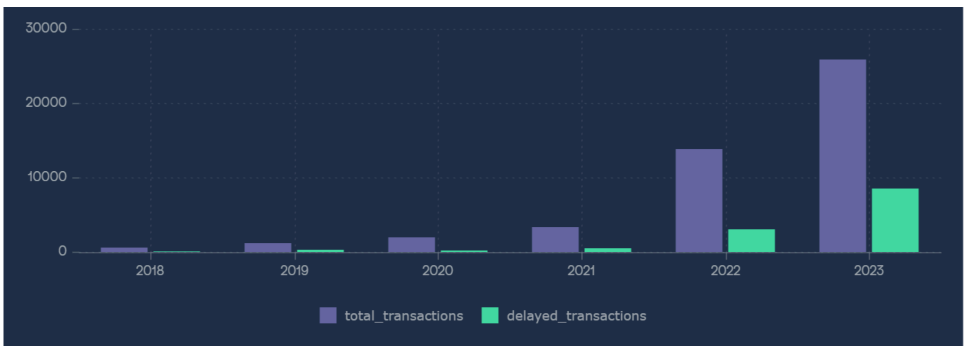

Figure 1 - Delay rate by year, 2018-2023

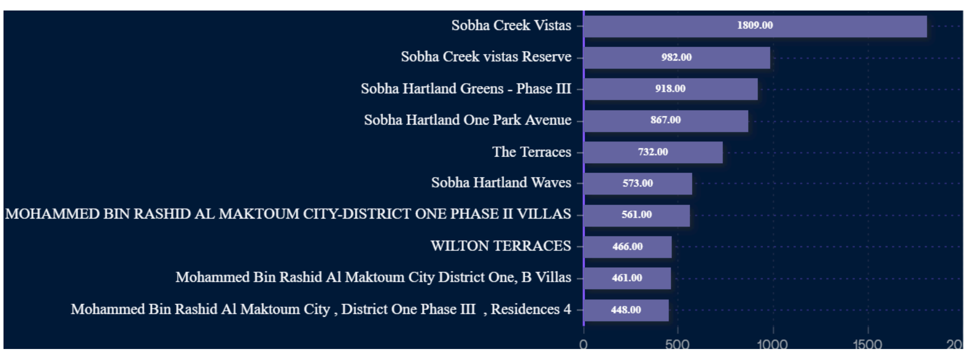

Figure 2 - Top 10 projects by delayed transactions

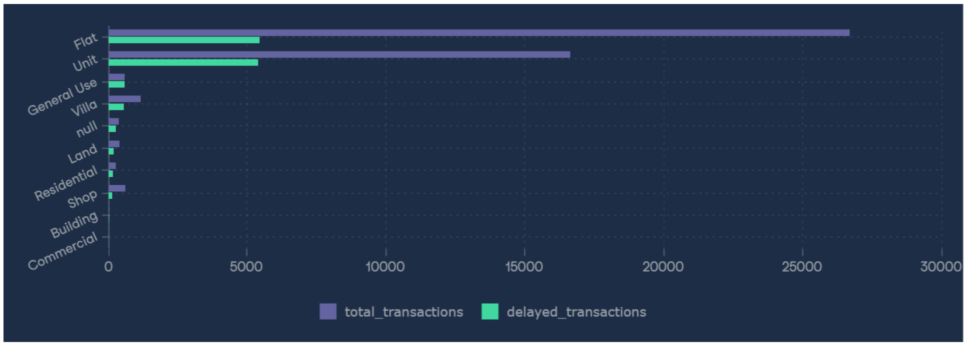

Figure 3 - Total vs delayed transactions by property type

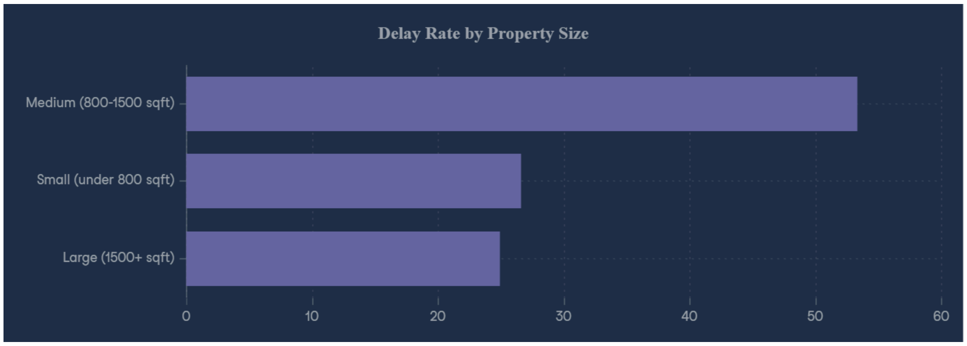

Figure 4 - Delay rate by property size

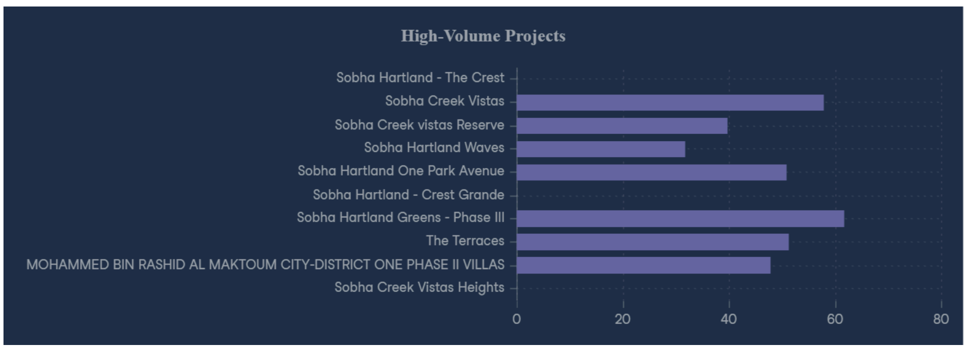

Figure 5 - Delay rate across high-volume projects

7. Key Findings

1. About 1 in 4 buyer transactions are delayed.

2. The delay rate went up as transaction volume went up, nearly tripling from 2018 to 2023.

3. Sobha Creek Vistas is the single biggest risk in the portfolio, both in count and in money.

4. General Use properties are delayed almost every time, 99.6%, even though it's a small category.

5. Some projects sell well but still have most of their revenue stuck in delays.

6. Medium-size properties (800-1500 sqft) are delayed more than small or large ones, 53.34%.

7. China accounts for a large number of transactions and a high delay rate, 54.35% of 2,769.

8. Delays aren't only about volume of Flats. Some of the largest single amounts of money, up to AED 95.6 million, are stuck in Villa and Land transactions.

8. Recommendations

1. Review Sobha Creek Vistas first. It has 1,809 delayed transactions and about AED 1.59 billion tied up, more than any other project.

2. Study what Sobha Hartland - The Crest is doing right. It has the highest sales volume in the portfolio and 0% delays, so it's a working example, not just a target to fix.

3. Check the approval and handover process for General Use units and Medium-size properties. Both show delay rates well above the rest of the portfolio.

4. Look at the paperwork and payment process for Chinese buyers. This is a large group of buyers with a delay rate above average.

5. Track high-value Villa and Land transactions separately. A few of these hold very large amounts of money and even one delay here has a bigger financial impact than several Flat delays.

9. Limitations

This analysis only uses the transaction status field. There's no expected or actual handover date in the data, so it can't confirm the real reason behind a delay, or that a delay means a late handover to the buyer. It also can't rule out other factors, like contractor issues or legal holds, since that data wasn't available. The findings describe patterns in the 2018-2023 data only.

10. Repository and How to Reproduce This

All SQL queries, the data dictionary, and the charts are on GitHub:

https://github.com/uroojfatimaanalyst/sobha-realty-delay-analysis

To reproduce this analysis

1. Get the cleaned CSV file (or clean the raw file following the steps in Section 3).

2. Load it into DataLab, or any SQL tool that supports DuckDB syntax, as a table named sobha_cleaned.csv.

3. Run the queries in sql/01_milestone1_queries.sql, then sql/02_milestone2_queries.sql, in order.

4. Each query's output is described next to it in the milestone reports in the Docs folder.

Repository structure

```
sobha-realty-delay-analysis/
├── README.md
├── data/
│   └── README_data_source.md
├── docs/
│   ├── Sobha_MP1_Report.pdf
│   └── Sobha_MP2_Report.pdf
├── sql/
│   ├── 01_milestone1_queries.sql
│   └── 02_milestone2_queries.sql
└── visuals/
    └── chart1.png ... chart7.png
```